

NABD

Echo_Feel



The Problem

The Communication Gap

Estimated Population: Between 4.5 to 5 million people.

Prevalence: Approximately 4.2% to 4.5% of the total Egyptian population



Social Isolation: Millions of deaf and hard-of-hearing individuals face daily isolation due to the "Communication Barrier" with people who do not understand sign language



Lack of Real-Time Interaction: Traditional methods like writing on paper or using mobile typing are slow, inefficient, and kill the flow of natural conversation.



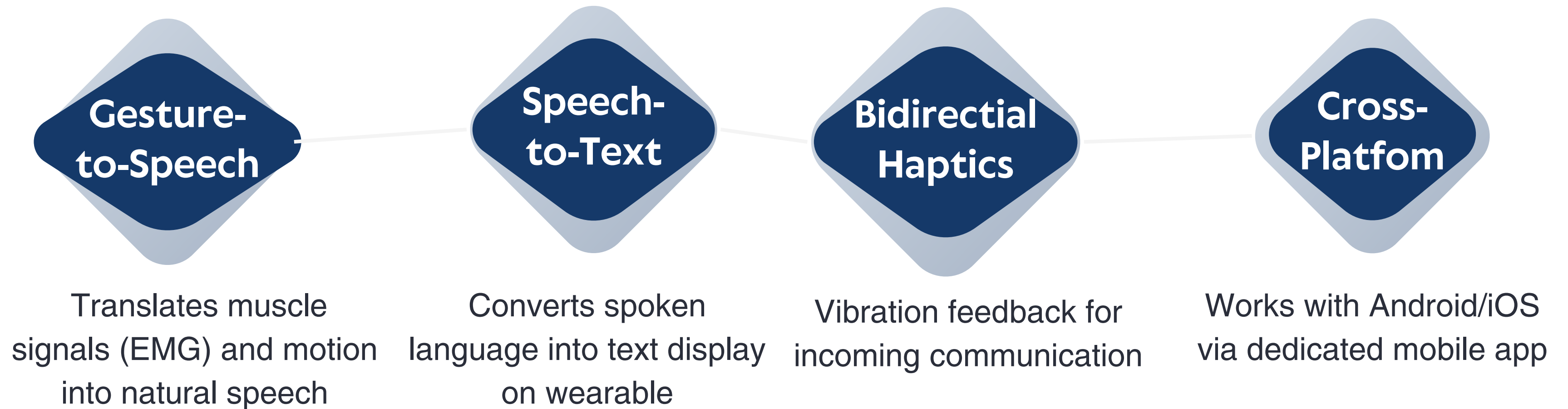
One-Sided Solutions: Most existing tools only translate sign language to text, but they don't help the deaf person "receive" or understand the spoken words of others in a seamless way.



Accessibility Challenges: Deaf individuals struggle to perform basic tasks in public places (hospitals, banks, or shops) because there is no immediate "translator" available 24/7.

Solution

Our Mission: To create a seamless, two-way communication bridge between the Deaf/Mute community and society using intelligent, scalable technology.



Product

SignSync is a wearable assistive technology ecosystem designed to bridge the communication gap between the Deaf/Mute community and the general public. Recognizing that over 60% of Sign Language gestures require two hands, SignSync utilizes a Dual-Wrist Architecture to capture full-body semantics. The system translates hand movements into spoken natural language and, conversely, converts spoken audio into visual text and haptic feedback pulses (conveying rhythm/intensity). It operates on a Hybrid AI Model: running locally on a smartphone for speed (Edge AI) and utilizing a subscription-based Cloud API for advanced processing and unlimited vocabulary.



Digital product

Dual-Mode Architecture :

Basic Tier (Free)

On-device Lite processing

Standard dictionary (100+ essential words/phrases)

Real-time speech-to-text display

Offline functionality

Haptic feedback control

Pro Tier (Subscription)

Cloud API connectivity

Unlimited vocabulary with contextual understanding

Multi-language support (Sign Language → English/Arabic/French/etc.)

Natural voice skins (gender, accent, age options)

Sentiment analysis for emotional context

Continuous AI model updates

Industry outlook

massive and growing market
opportunity

1+ BILLION

people globally live with
significant disability

466 BILLION

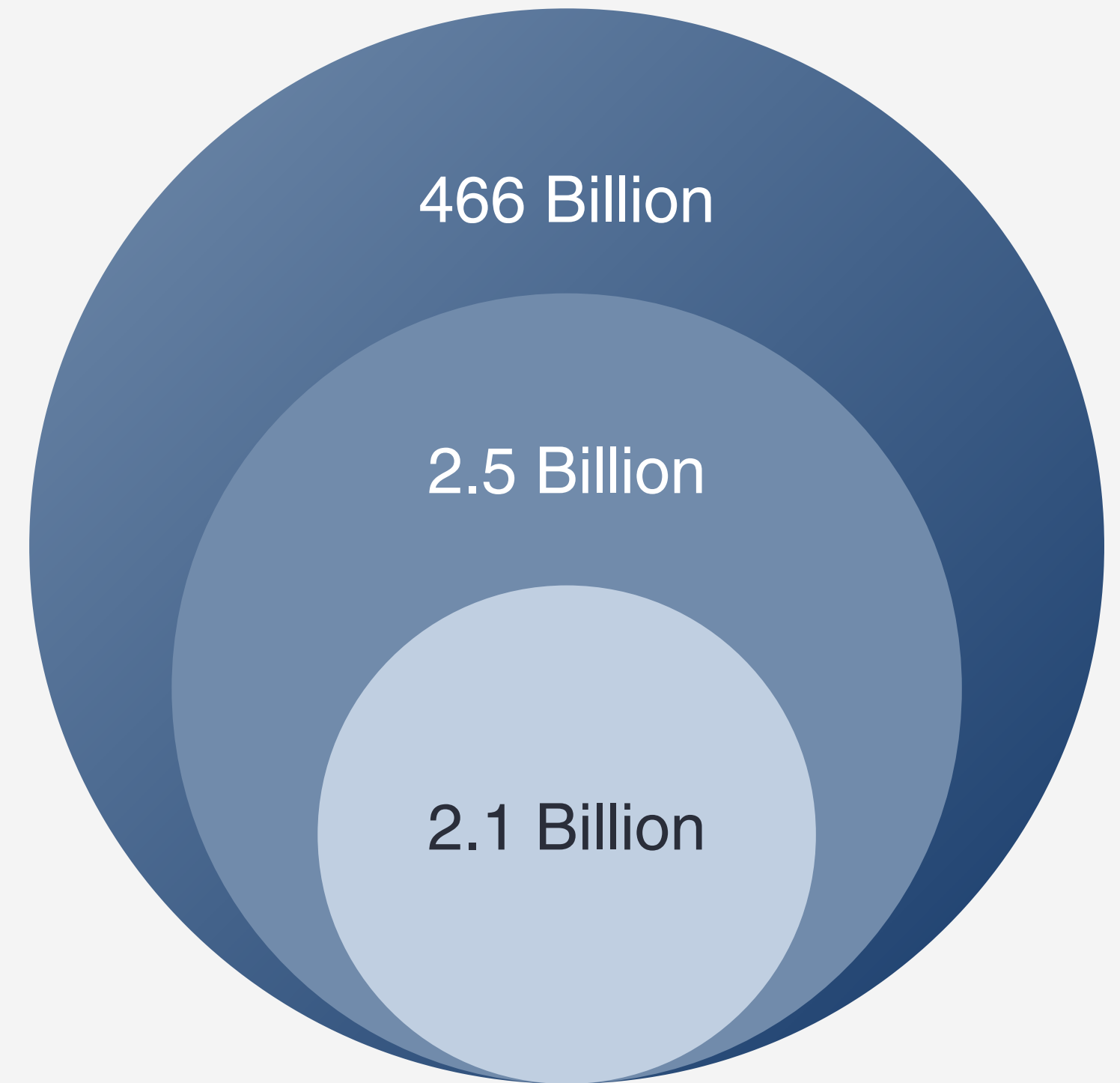
worldwide suffer from
disabling hearing loss

2.5 BILLION

projected to
experience hearing loss
by 2050

2.1 BILLION

people over 60 by 2050
(double today's number)



KEY PARTNERS



- Cloud service providers for AI hosting and scalable computing
- Rehabilitation centers and special education schools for pilot testing and user feedback
- NGOs and disability support organizations for community access and outreach
- Universities and research institutions for AI and hardware development
- Electronics manufacturers and suppliers for device production
- Healthcare and government entities for large-scale deployment (future)

KEY ACTIVITIES



- Designing and manufacturing the wearable device
- Developing and optimizing AI models
- Training AI with real user data
- Maintaining the mobile app and cloud services
- User testing and experience improvement
- Marketing and partnership development

KEY RESOURCES



- Hardware components (ESP32, EMG, IMU sensors)
- Mobile application and software platform
- AI models and training datasets
- Cloud infrastructure and servers
- Skilled development and research team

VALUE PROPOSITIONS



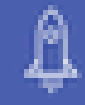
- Real-time translation of muscle signals and gestures into natural speech
- Speech-to-text and haptic feedback for two-way communication
- Works offline using local AI processing
- Higher accuracy and advanced understanding through cloud AI
- Wearable, discreet, and user-friendly design
- Multi-language support and continuous AI updates
- A practical, daily-use communication solution

CUSTOMER RELATIONSHIP



- Self-service through the mobile application
- In-app customer support and tutorials
- Continuous software and AI updates
- User community for feedback and improvement
- Free Basic plan and premium subscription model

CHANNELS



- Mobile application (Android / iOS)
- Online device sales (e-commerce platforms)
- Partnerships with rehabilitation centers and schools
- Social media and digital marketing campaigns
- Technology exhibitions and startup events

CUSTOMER SEGMENTS



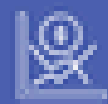
- Deaf and mute individuals
- People with hearing impairments
- Families and caregivers of deaf/mute users
- Special education schools and rehabilitation centers
- NGOs supporting people with disabilities
- Government and educational institutions(future expansion)

COST STRUCTURE



- Hardware manufacturing costs (ESP32, EMG and IMU sensors, display, vibration motor, casing, and assembly)
- Research and development costs for AI model training and optimization
- Mobile application development and maintenance costs
- Cloud infrastructure costs for AI processing, storage, and API usage (Pro users only)
- Marketing and user acquisition costs
- Customer support and operational costs

REVENUE STREAM



- Hardware Sales (One-Time Purchase):
 - Selling the NeuroVoice wearable device at an affordable price to ensure accessibility.
- Monthly Subscription (Pro Plan):
 - Recurring monthly payments for cloud-based AI features such as unlimited vocabulary, smart prediction, voice customization, and multi-language translation.
- Institutional Licensing & Bulk Subscriptions:
 - Annual contracts with schools, rehabilitation centers, and NGOs for bulk devices and centralized cloud access.
- Government & Healthcare Partnerships:
 - Long-term service contracts with public institutions to deploy NeuroVoice as an assistive technology solution.
- Data-Driven AI Improvement (Ethical & Anonymous):
 - Aggregated and anonymized usage data used to improve AI models and offer enterprise-level insights (without selling personal data).

SDGs Impact Distribution



SDG 10 – Reduced Inequalities
Enables deaf and mute individuals to communicate equally with society using AI-powered assistive technology.



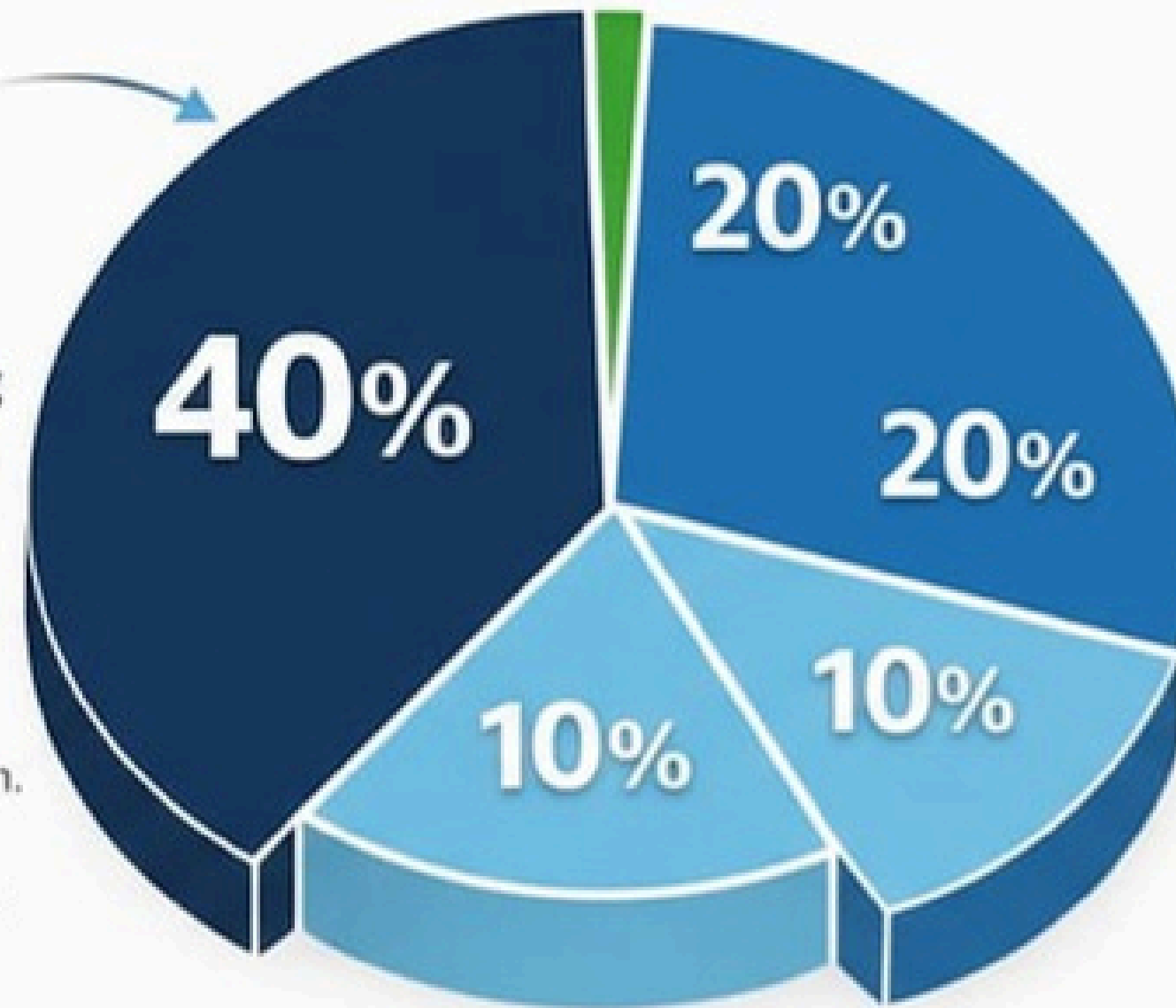
SDG 3 – Good Health & Well-being
Reduces social isolation and improves mental well-being.



SDG 4 – Quality Education
Supports inclusive education by improving classroom communication.



SDG 8 – Decent Work & Economic Growth
Helps users access job opportunities by breaking communication barriers.



Focus on Social Inclusion & Reduced Inequalities.

Compatitor analysis



Company	Approach	Strengths	Weaknesses
Glovatrix (India)	AI smart glove, sign-to-speech	Funded startup (2025), multiple languages	Hardware-dependent, no hybrid cloud tier
Control Bionics	Assistive devices, speech-impaired	Established player	Traditional hardware focus
Tobii Dynavox	Eye-tracking, AAC devices	Market leader, clinical validation	High cost, limited sign language
Voiceitt	Non-standard speech recognition	\$5.39M funding, ML focus	Speech-only, no sign language
SignAll	Sign language translation	Computer vision based	Camera-dependent, not wearable

Growth strategy

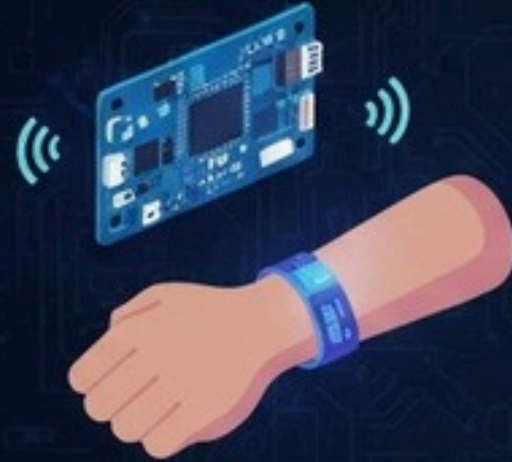
Product development roadmap

- ◆ Phase 1: hardware foundation
- ◆ Phase 2: mobile app MVP
- ◆ Phase 3: cloud integration
- ◆ Phase 4: user training
- ◆ Phase 5: market launching

PROJECT TIMELINE

Phase 1: Hardware Foundation

Q3 2026



Phase 3: Mobile App MVP

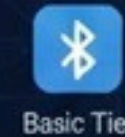
Q1 2026



Basic Tier with
Tenstaforiwo Lite,
100+ words

Phase 5: Market Launch

Q3 2026



Basic Tier



Pro Tier

Phase 2: Mobile App MVP

Q3 2026



ESP32 + EMG/IMU assembly,
BLE connectivity

Phase 4: User Training

Q1 2027



Model fine-tuning with
real users

Phase 5: Market Launch

Q2 2027



Model
fine-tuning

Commercial relasial
release with both tiers

REQUIRED FUNDING

total ask: 50\$

Component	Function	Qty	Unit Price (EGP)	Total (EGP)	Note
ESP32 (Dev Board)	Microcontroller	2	300	600	WiFi/BT included.
MPU6050	Gyro + Accelerometer	2	120	240	Motion tracking.
AD8232 (Module)	EMG Sensor Substitute	2	400	800	<i>Critical Optimization: Using ECG sensor hacked for EMG to save ~3000 EGP.</i>
OLED Display	0.96" I2C Screen	1	150	150	Right wrist only.
Li-Ion Battery	Power (18650 type)	2	110	220	Or salvage from old laptops.
TP4056	Charging Module	2	35	70	For USB-C charging.
MAX98357A	I2S Audio Amp	1	120	120	Digital Audio output.
Micro Speaker	Sound Output	1	50	50	3 Watt / 4 Ohm.
Vibration Motor	Haptic Feedback	1	30	30	Coin type.
Consumables	3D Print / Wires / Straps	-	-	300	Estimate.
TOTAL				**~ 2,580 EGP**	(Approx. \$50 USD)



SWOT ANALYSIS



STRENGTHS

- Hybrid Architecture
- **TWO-WAY COMMUNICATION**
- **SUBSCRIPTION MODEL**
- **SCALABLE TECHNOLOGY**

THREATS

- **GROWING MARKET (\$29B)**
- **AGING POPULATION**
- **GOVERNMENT MANDATES**
- **PARTNERSHIPS**

WEAKNESSES

- **HARDWARE DEPENDENCY**
- **BATTERY LIFE**
- **USER TRAINING NEEDED**
- **INTERNET DEPENDENCY**

OPPORTUNITIES

- **BIG TECH COMPETITION**
- **CLOUD COST RISKS**
- **ACCURACY EXPECTATIONS**
- **PRIVACY REGULATIONS**



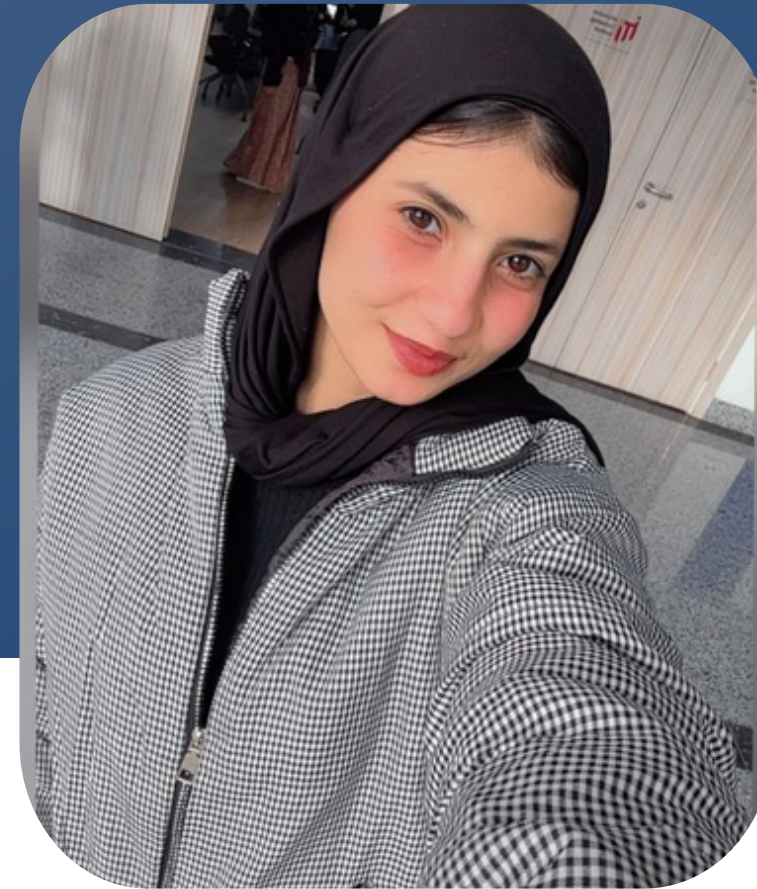
MEET OUR TEAM



Nada Ebrahim



Basmala Emad



Kenzy Ahmed



Abdalrahman Othman

Connect with us.



Email
abdalrahman.hossam.othman@gmail.com



Social Media
abdalrahman-othman-



Call us
01029388659



THANK YOU

